Medicare Part D Utilization and Cost Modeling

Giorgos Tzimas

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Abstract

This project analyzes Medicare Part D prescription drug utilization and cost using public CMS geography-by-drug files from 2019 through 2024. The analysis focuses on state-drug records for the 50 states and District of Columbia, with the goal of understanding cost concentration, testing whether selected drug flags are associated with beneficiary-level cost, and evaluating whether prior-year utilization and cost features can identify high-cost records in the following year. The analytic file contained 524,753 state-drug-year records after excluding suppressed or unusable cost, beneficiary, claim, and fill values. In 2024, represented state-level drug cost totaled $280.8 billion, and the top 5% of state-drug records accounted for 74.5% of represented drug cost. A log-cost regression found that long-acting opioid, antibiotic, opioid, and antipsychotic flags were statistically associated with cost per beneficiary after adjustment for volume, age 65+ share, cost-sharing share, and year. For prediction, a simple current-cost baseline, logistic regression, and random forest all performed well when identifying 2024 high-cost records. The result is less a story about model complexity than about cost persistence, data quality, and the value of transparent review lists for pharmacy cost monitoring.

1 Introduction

Prescription drug spending is a large and visible part of the U.S. health care cost problem, and Medicare Part D is one of the main public programs for prescription drug coverage among older adults and disabled beneficiaries. CMS publishes National Health Expenditure data and Part D public-use files that make it possible to study drug cost and utilization patterns without using protected patient-level records (Centers for Medicare & Medicaid Services, 2026c,a). The Part D Prescribers by Geography and Drug files are especially useful for portfolio-scale analysis because they summarize prescription claims by geography and drug, including prescriptions, 30-day fills, total drug cost, beneficiary counts, low-income subsidy cost-sharing measures, and selected drug flags (Centers for Medicare & Medicaid Services, 2025, 2026b).

Health care cost data are rarely tidy in the statistical sense. They tend to be right-skewed, concentrated in relatively small subgroups, and sensitive to how outcomes are transformed or modeled (Diehr et al., 1999; Manning and Mullahy, 2001; Buntin and Zaslavsky, 2004; Mihaylova et al., 2011). That matters in pharmacy analytics because a small set of state-drug combinations can dominate spending even when common lower-cost medications represent most records. A useful analysis therefore needs to do more than train a model. It needs to measure data quality, show where cost is concentrated, benchmark models against simple baselines, and state clearly what can and cannot be inferred.

This project treats each state-drug-year record as a unit of analysis. The records are not individual beneficiaries or raw claim lines, so the findings should be read as utilization and cost-pattern evidence rather than clinical causal evidence. The main objective is practical: build a clean workflow that can summarize cost trends, test prespecified hypotheses, and produce review lists that a pharmacy or healthcare analytics team could inspect further.

2 Research Questions and Hypotheses

The analysis was organized around three questions.

H1. Drug flags and cost per beneficiary. After adjustment for utilization volume, year, age 65+ share, and beneficiary cost-sharing share, CMS drug flags for opioids, long-acting opioids, antibiotics, and antipsychotics are associated with cost per beneficiary.

H2. High-cost prediction. Prior-year state-drug utilization and cost features can identify state-drug records that fall in the top decile of next-year cost per beneficiary. Because drug costs tend to persist across years, the predictive models should also be compared with a current-cost baseline.

H3. Review scenario. A targeted review list based on predicted high-cost records can concentrate enough represented drug cost that a modest avoidable-cost reduction would offset review-program expense. This is a planning scenario, not an estimate of causal savings.

3 Data

The data source was the CMS Medicare Part D Prescribers by Geography and Drug public-use files for 2019 through 2024 (Centers for Medicare & Medicaid Services, 2026a). The files are based on Medicare administrative claims for beneficiaries enrolled in Part D and include final-action prescription drug claims summarized by geography and drug (Centers for Medicare & Medicaid Services, 2026b). The analysis used state-level records for the 50 states and District of Columbia. National rows and rows with missing state names were excluded from the analytic file.

CMS suppresses some values to protect privacy and reduce disclosure risk. In the downloaded CSV files, suppressed cells appear as symbols such as `*` or `#`; these were parsed as missing values rather than zeros. Records needed positive total drug cost, total beneficiaries, claims, and 30-day fills to enter the main analytic file. Age 65+ measures and cost-sharing fields were retained when available and imputed with training-set medians for the prediction task.

Table 1: Data quality and cohort construction summary.

Measure	Value
Raw rows	693,939
State rows for 50 states plus DC	646,117
Analytic rows with positive core fields	524,753
Years	6
Distinct state codes	51
Distinct generic names	1,916
Distinct brand names	3,551
Suppressed or missing total beneficiary values	130,194
Suppressed or missing age 65+ beneficiary values	292,686
Suppressed or missing age 65+ cost values	126,456

The beneficiary counts in these files are drug-record counts, not unique member counts across all medications. A beneficiary using multiple drugs can contribute to multiple state-drug records.

For that reason, outcomes such as cost per beneficiary should be interpreted as cost per beneficiary record within a drug-geography combination.

4 Methods

4.1 Feature Engineering

For each state-drug-year record, the workflow calculated cost per beneficiary, cost per claim, cost per 30-day fill, claims and fills per beneficiary record, prescribers per 100 beneficiary records, age 65+ beneficiary and cost shares, beneficiary cost-sharing share, and binary indicators for opioid, long-acting opioid, antibiotic, and antipsychotic flags.

The analysis was written in R using tidyverse tools for data preparation and plotting (R Core Team, 2025; Wickham et al., 2019). The random forest model used the `ranger` implementation (Breiman, 2001; Wright and Ziegler, 2017), and ROC curves were evaluated with `pROC` (Robin et al., 2011).

4.2 Statistical Analysis

For H1, the outcome was log-transformed cost per beneficiary:

$$\log(1 + \text{cost per beneficiary})$$

The regression adjusted for year, log total beneficiaries, log 30-day fills, fills per beneficiary, age 65+ beneficiary share, beneficiary cost-sharing share, and the four CMS drug flags. Coefficients were exponentiated and interpreted as approximate percent differences in cost per beneficiary. The model was meant to describe association, not causal effects.

For H2, the panel was arranged so year t features predicted whether the same state-drug record would be in the top decile of cost per beneficiary in year $t + 1$. The training window used 2019 to 2022 predictors with 2020 to 2023 targets. The test window used 2023 predictors with 2024 targets. Logistic regression, random forest classification, and a current-cost baseline were compared using ROC AUC, average precision, precision and recall in the top 10% review list, and the share of represented 2024 drug cost captured by that review list.

For H3, the random forest top 10% review list was used in a simple planning scenario. The scenario assumed a review cost of $2,000 per state-drug pair and evaluated avoidable-cost reductions from 0.5% through 5.0%. The scenario does not assume that all high-cost spending is waste. It is a way to show the reduction needed for a review process to pay for itself.

5 Results

5.1 Annual Cost and Utilization

Across the analytic records, represented state-level Part D drug cost increased from $178.8 billion in 2019 to $280.8 billion in 2024. Total claims and 30-day fills also increased over the period, while the median cost per beneficiary record declined in 2024. That combination suggests a broad increase in represented volume and total cost, with median record-level cost moving differently from aggregate spending.

Table 2: Annual summary of analytic state-drug records.

Year	Rows	Total drug cost	Total claims	30-day fills	Median cost/beneficiary
2019	87,346	$178.8B	1,473,296,055	2,494,202,873	$524
2020	86,886	$193.5B	1,467,737,709	2,592,620,434	$540
2021	86,548	$210.0B	1,469,947,090	2,656,433,432	$531
2022	87,120	$234.0B	1,513,061,140	2,760,953,846	$536
2023	87,782	$268.7B	1,585,596,314	2,902,154,978	$532
2024	89,071	$280.8B	1,678,960,791	3,092,954,076	$449

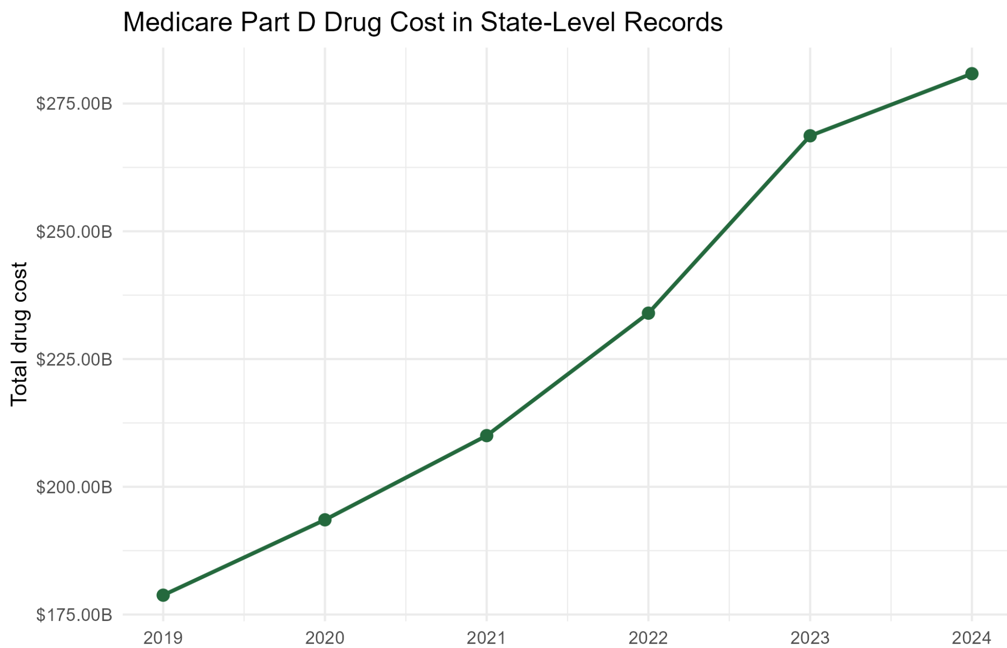

Figure 1: Total represented drug cost in analytic state-drug records, 2019–2024.

5.2 Cost Concentration

The 2024 distribution was highly concentrated. The top 5% of state-drug records accounted for 74.5% of represented drug cost. This pattern is typical for health care cost data and is one reason that simple averages can hide the operational problem (Diehr et al., 1999; Mihaylova et al., 2011).

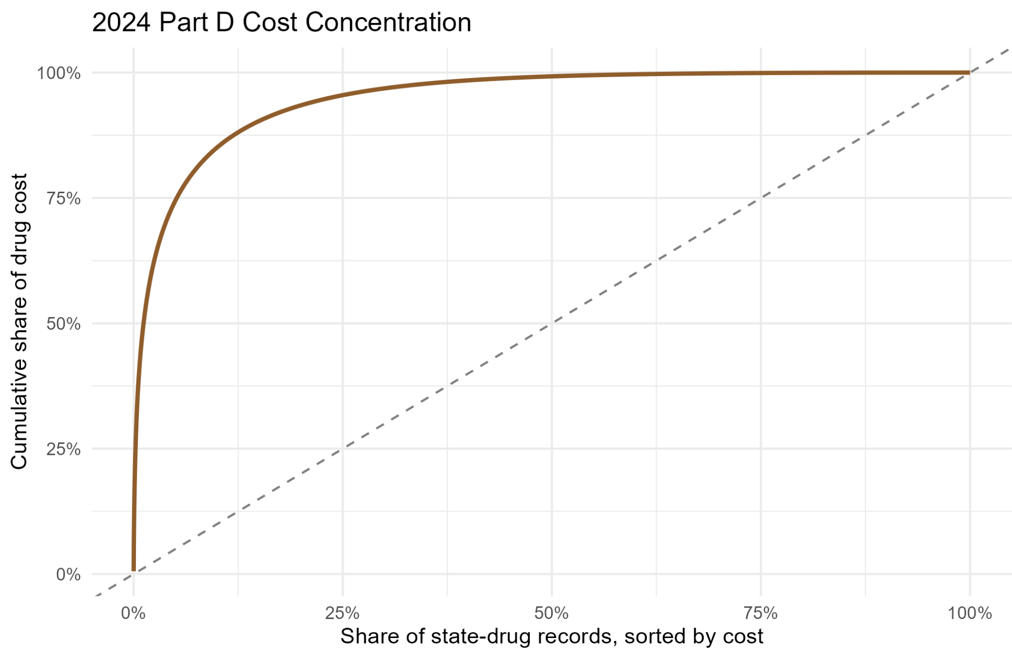

Figure 2: Cost concentration among 2024 state-drug records.

The highest-cost 2024 drug records were dominated by anticoagulants, diabetes medications, respiratory medications, immunology products, and oncology-related therapies. This table is descriptive; it does not imply that these medications are inappropriate or avoidable.

Table 3: Top 10 2024 drugs by represented state-level drug cost.

Generic	Brand	Total drug cost	Median cost/beneficiary
Apixaban	Eliquis	$20.6B	$4,366
Semaglutide	Ozempic	$12.9B	$6,478
Empagliflozin	Jardiance	$11.3B	$4,357
Tirzepatide	Mounjaro	$6.3B	$6,803
Rivaroxaban	Xarelto	$6.2B	$4,490
Dulaglutide	Trulicity	$5.4B	$6,614
Fluticasone/Umeclidin/Vilanter	Trelegy Ellipta	$5.3B	$4,089
Dapagliflozin Propanediol	Farxiga	$5.1B	$4,046
Adalimumab	Humira(Cf) Pen	$4.2B	$64,532
Lenalidomide	Revlimid	$4.1B	$96,369

5.3 H1: Drug Flags and Cost per Beneficiary

After adjustment, long-acting opioid records had an estimated 59.3% higher cost per beneficiary than otherwise similar non-flagged records. Antibiotic-flagged records were 35.9% higher, opioid-flagged records were 5.2% higher, and antipsychotic-flagged records were 5.6% lower. The confidence intervals for these flag terms did not cross zero.

Table 4: Adjusted associations between CMS drug flags and cost per beneficiary.

Flag	Percent difference	95% CI low	95% CI high	p-value
Opioid	5.2%	1.4%	9.1%	0.007
Long-acting opioid	59.3%	50.8%	68.2%	<0.001
Antibiotic	35.9%	33.2%	38.6%	<0.001
Antipsychotic	-5.6%	-8.0%	-3.2%	<0.001

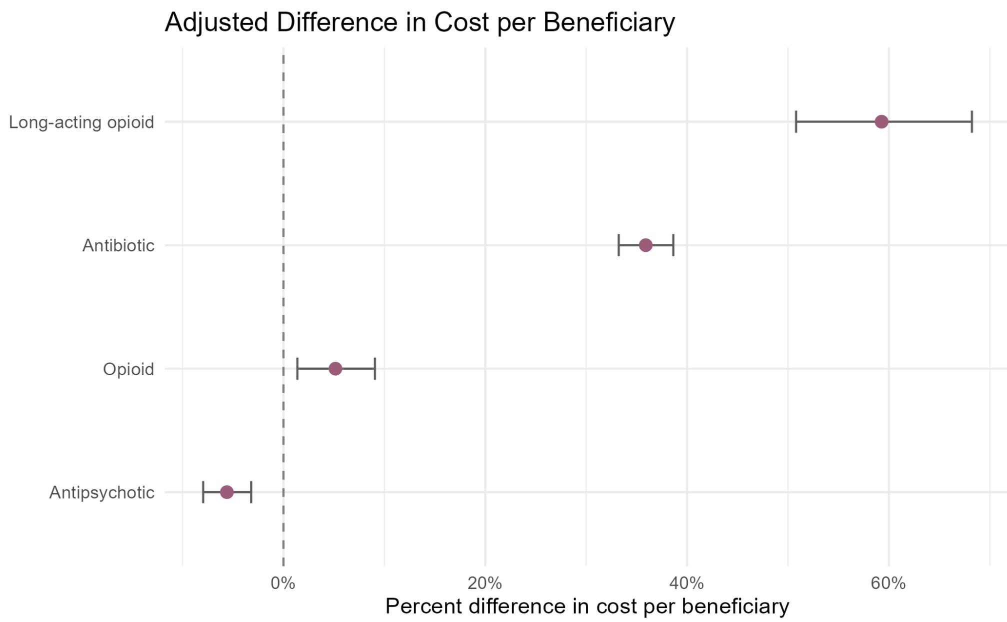

Figure 3: Adjusted percent differences in cost per beneficiary for selected CMS drug flags.

These differences should not be read as direct drug-class effects. The CMS flags are broad indicators, and the regression does not control for diagnosis, severity, plan design, rebates, or formulary management. The finding is still useful because it shows that flagged categories carry distinct cost profiles even after basic volume adjustment.

5.4 H2: High-Cost Prediction

The 2024 high-cost target was very predictable, but the strongest lesson was persistence rather than model complexity. The current-cost baseline had ROC AUC 0.999 and average precision 0.990. Logistic regression and random forest performed similarly, with random forest ROC AUC 0.999 and average precision 0.981.

Table 5: Prediction performance for 2024 high-cost state-drug records.

Model	ROC AUC	Avg. precision	Precision	Recall	Cost share captured
Logistic regression	0.999	0.989	90.7%	98.8%	37.2%
Random forest	0.999	0.981	90.5%	98.6%	37.2%
Current-cost baseline	0.999	0.990	90.7%	98.9%	37.1%

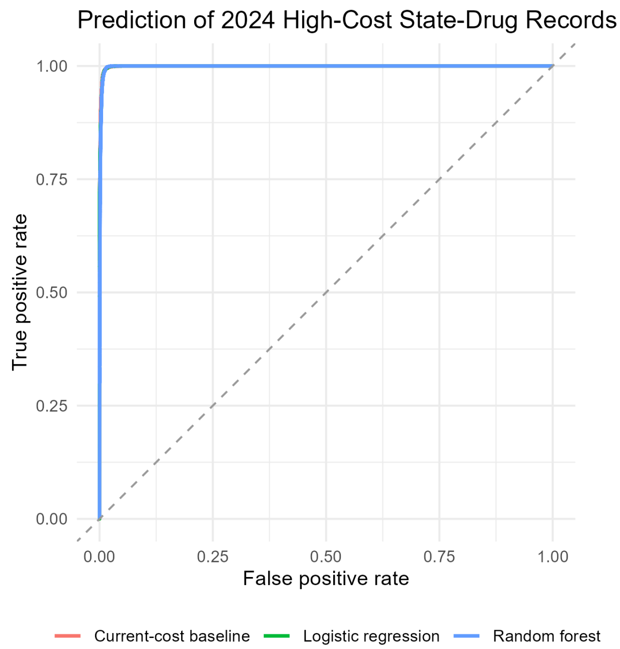

Figure 4: ROC curves for the 2024 high-cost prediction task.

The top random forest risk decile contained a 90.5% high-cost rate and captured 37.2% of represented 2024 drug cost. Lower risk deciles had almost no high-cost records by the top-decile definition, although some lower-risk deciles still contained meaningful total spending because total drug cost and cost per beneficiary are related but not identical.

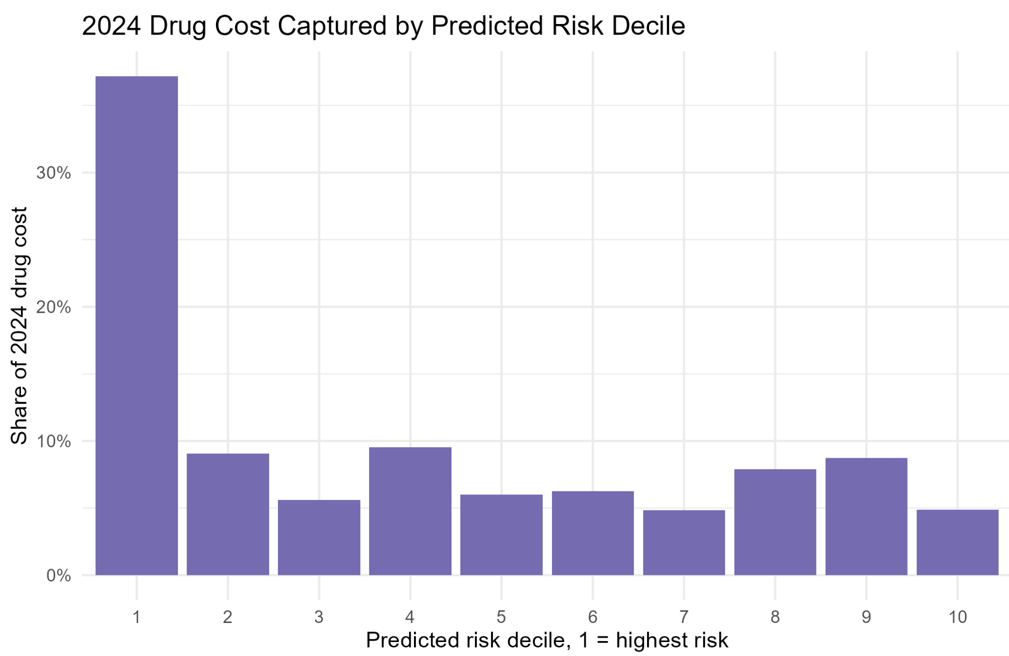

Figure 5: Share of represented 2024 drug cost captured by random forest risk decile.

5.5 H3: Review Scenario

The random forest top 10% review list included 8,282 state-drug pairs and represented $102.8 billion in 2024 drug cost. At an assumed review cost of $2,000 per state-drug pair, the review program would cost $16.6 million. The break-even avoidable-cost reduction was 0.02%.

Table 6: Planning scenario for a targeted pharmacy cost review list.

Avoidable reduction	Program cost	Targeted drug cost	Avoidable cost	Net savings
0.5%	$16.6M	$102.8B	$514.0M	$497.4M
1.0%	$16.6M	$102.8B	$1.0B	$1.0B
2.0%	$16.6M	$102.8B	$2.1B	$2.0B
3.0%	$16.6M	$102.8B	$3.1B	$3.1B
5.0%	$16.6M	$102.8B	$5.1B	$5.1B

This scenario is intentionally modest in structure. It does not claim that a review process will produce savings. It says that the review list concentrates enough represented spending that the reduction needed to cover review expense is small. Whether that reduction is achievable would require clinical review, plan and formulary context, and a stronger evaluation design.

6 Discussion

The analysis supports three practical conclusions.

First, state-level Part D spending is heavily concentrated. That concentration means a broad, undifferentiated cost-control strategy is unlikely to be efficient. Review programs, dashboards, or outreach workflows should prioritize the highest-cost state-drug combinations and then separate clinically necessary spending from opportunities such as duplicate therapy, avoidable utilization, site-of-care issues, formulary alternatives, adherence gaps, or coding and data-quality problems.

Second, the predictive modeling results are a reminder that simple baselines matter. Current cost per beneficiary was nearly as strong as the logistic regression and random forest models. That does not make modeling useless. It means the main analytical value is in building a reliable monitoring workflow, validating how stable the high-cost target is, and producing review lists with transparent features. In this setting, an analyst should be careful about presenting a complex model as if it discovered a hidden pattern when a simpler benchmark explains most of the signal.

Third, the project highlights why data quality needs to sit next to modeling. CMS suppression rules affected beneficiary and age 65+ fields, and those missing values were not random artifacts of a spreadsheet. They came from a disclosure policy. A healthcare analytics workflow has to treat those values as part of the data-generating process and communicate the effect on the analysis.

7 Limitations

The main limitation is aggregation. The files summarize Part D claims by geography and drug; they do not include patient-level enrollment, diagnoses, lab results, provider contracts, plan benefit design, rebates, or clinical outcomes. Cost per beneficiary is calculated within a state-drug record and does not represent unique patient-level annual cost.

The analysis also does not identify appropriate or inappropriate utilization. High cost can reflect clinically necessary care, expensive specialty therapy, disease severity, drug mix, or plan and market factors that are not visible in this file. The review scenario is therefore a business-planning exercise rather than an estimate of actual savings. A stronger evaluation would pair the review list with clinical criteria, pharmacy benefit data, and a pre-post or matched comparison design.

8 Conclusion

Public CMS Part D data can support a useful healthcare analytics workflow when the analysis is honest about aggregation, suppression, and cost persistence. In this project, represented 2024 drug cost was highly concentrated, selected CMS drug flags were associated with cost per beneficiary, and next-year high-cost records were identifiable with both simple and model-based approaches. The strongest practical takeaway is that transparent cost monitoring, careful benchmarking, and disciplined review-list design may be more valuable than model complexity by itself.

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